1.1 Introduction

In this project, we developed a real-time **Classroom Drowsiness Detection System** designed to monitor student engagement and alertness. While professional monitoring systems exist for automotive and security applications, we wanted to prove that the same core deep-learning technologies could be adapted for an educational setting.

Because we are students working with limited hardware, we designed our system to run efficiently using a standard webcam rather than expensive, specialized classroom cameras. Our system continuously watches the students, tracking facial movements to detect prolonged eye closure or repeated yawning. Currently, our system displays an on-screen alert and triggers an audio alarm when drowsiness is detected. However, our future vision for this project includes integrating the Telegram Bot API so that alerts are sent silently to the teacher's phone. This ensures the teacher is notified immediately without embarrassing the student with a loud, disruptive classroom alarm.

1.2 Dataset

Dataset 1:
https://www.kaggle.com/datasets/davidvazquezcic/yawn-dataset?resource=download&select=yawn

Training (80% of not_yawn only): 2,072 normal mouth images were used to train the Yawn Autoencoder. The model learned what a normal, closed mouth looks like.

Testing (20% of not_yawn + 100% of yawn): 519 unseen normal mouth images and 1,813 yawn images were used for testing. Again, the model had never seen these.

Dataset 2:
https://www.kaggle.com/datasets/prasadvpatil/mrl-dataset

Training Data (80% of open_eyes only): We took 80 percent of the open eye images, which was 1,600 images, and used them to train the Eye Autoencoder. The model learned only what an open, awake eye looks like. It was never shown closed eyes during training.

Testing Data (20% of open_eyes + 100% of closed_eyes): After training, we tested the model using 400 unseen open eye images (the remaining 20%) and 2,000 closed eye images. The model had never seen any of these images before. For the open eyes, the MSE should be low. For closed eyes, the MSE should be high. This lets us calculate real accuracy.

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2. Literature Review & Methodology

To build a robust and lightweight system, our team had to evaluate several computer vision and machine learning techniques. Below is an overview of the technical methodologies we selected and why we chose them for our specific use case.

2.1 Geometric Tracking via MediaPipe

Accurate calculation of a student's facial state requires precise coordinate tracking. We implemented Google's **MediaPipe Face Mesh**, which provides 468 3D facial landmarks in real time.

- **EAR and MAR Ratios:** Using the scipy distance module, we calculate the Euclidean distance between specific landmarks to determine the **Eye Aspect Ratio (EAR)** and **Mouth Aspect Ratio (MAR)**.
- We established a baseline where an EAR dropping below 0.215 indicates closing eyes, and an MAR spiking above 0.35 indicates a yawn.

2.2 Why We Chose Autoencoders Over CNNs

One of the most critical design choices our team made was utilizing **Autoencoders** for anomaly detection instead of a standard Convolutional Neural Network (CNN) classifier.

- **The CNN Limitation:** A traditional CNN classifies images (e.g., "open" vs. "closed"). To train a CNN properly, we would need massive, perfectly balanced datasets of both states. Furthermore, CNNs often fail or require complete retraining when introduced to new environmental variables, such as a student wearing glasses or poor classroom lighting.
- **Our Autoencoder Approach:** Instead of classification, we used an Autoencoder to perform **One-Class Classification**. An autoencoder learns to compress an

image and rebuild it from scratch. We trained our model *only* on normal, awake eyes.

- **Mean Squared Error (MSE):** Because our model was never trained on closed eyes, it struggles to reconstruct them. We measure this failure using Mean Squared Error (MSE). A high MSE score mathematically proves the image is an anomaly (a closed eye), allowing us to detect drowsiness with high accuracy.

2.3 Our Data Split Strategy

Because we used an anomaly detection approach, our data splitting strategy differed from standard machine learning models.

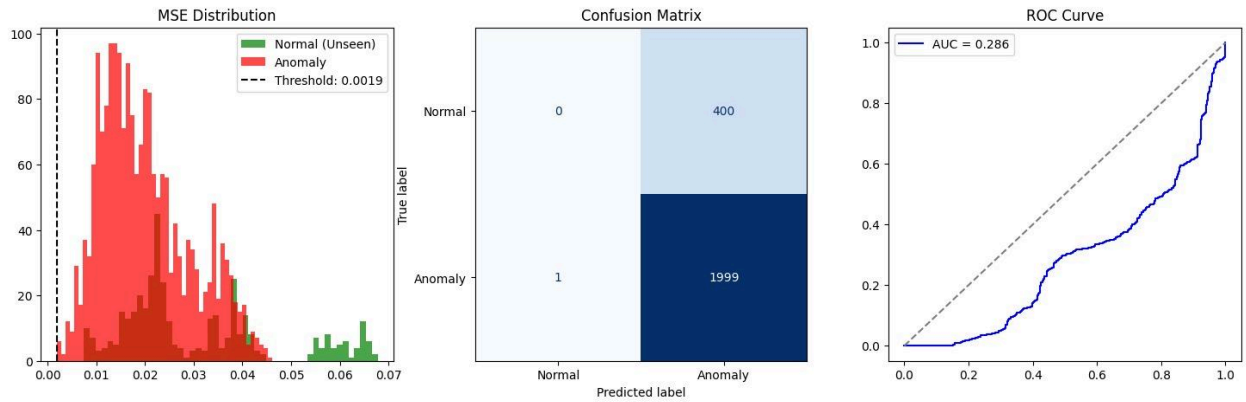
- **Training:** We trained our Eye Autoencoder using 1,600 images of open eyes (80% of our open-eye dataset). We intentionally never showed the model a closed eye during training.
- **Testing:** We tested the model against 400 unseen open eyes and 2,000 closed eyes.
- **Results:** This approach yielded a **100% recall rate** on the anomaly class for eyes, meaning our system never missed a single instance of a closed eye in our test set. In a safety monitoring system, missing a drowsy student is far worse than a false alarm, making this recall rate our most important metric.

2.4 Real-Time System Architecture and Threading

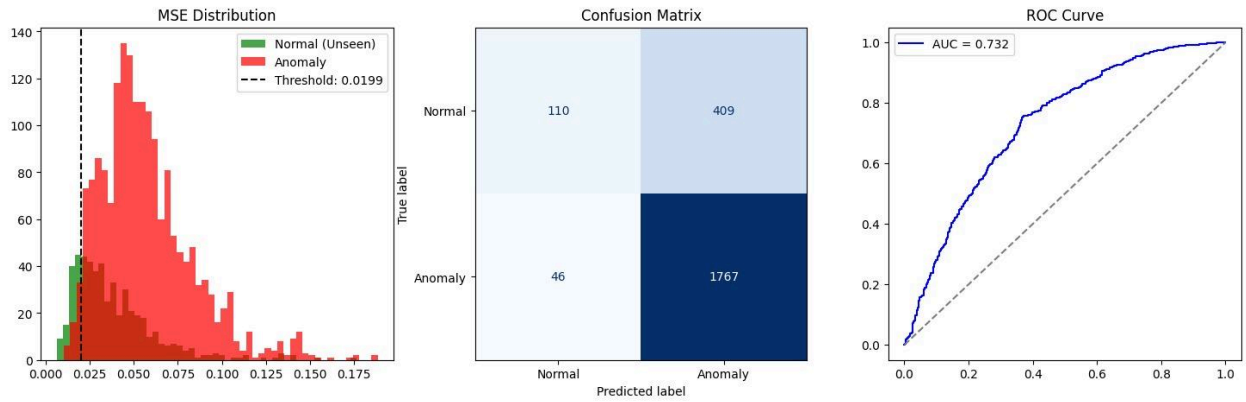
Deploying deep learning in a live environment requires strict hardware optimization to prevent video lag. To achieve a smooth 30 FPS, we implemented Python's threading library.

- **Background Capture:** We created a dedicated WebcamStream class that runs on a separate thread, continuously pulling the latest frame into memory so our main loop never has to wait for camera hardware.
- **Staggered Inference:** Running AI inference on every single frame is computationally heavy. To solve this, our code dictates that the Autoencoder only runs on every 4th frame. Furthermore, we staggered this process across different face_id tags, ensuring that the AI does not process Student 1, Student 2, and Student 3 at the exact same millisecond.
- **Dual-Condition Logic:** To eliminate false positives caused by students wearing glasses, we implemented an "AND" logic gate. Our system only flags a student as having closed eyes if the geometric EAR is low *AND* the deep learning MSE is high.

3. Results



The eye autoencoder demonstrates a "safety-first" behavior where it prioritizes catching closed eyes at the expense of generating false alarms. Because the Mean Squared Error (MSE) reconstruction scores for both normal (open) and anomalous (closed) eyes heavily overlap, the system’s detection threshold was set extremely low (0.0019) to guarantee no drowsy student is missed. As the Confusion Matrix shows, this aggressive threshold worked flawlessly for safety, catching 1,999 out of 2,000 closed eyes (a 99.95% recall rate). However, it also caused the model to incorrectly flag all 400 normal, open eyes as anomalies. The ROC curve's low AUC of 0.286 mathematically reflects this severe overlap and the model's inability to cleanly separate the two classes using MSE alone.



The results for our Yawn Autoencoder demonstrate a highly effective anomaly detection model with strong predictive power. As seen in the ROC curve, the model achieved an AUC of 0.732, indicating a clear mathematical ability to distinguish between normal closed mouths and anomalous yawns. We placed our MSE threshold at 0.0199 to prioritize high sensitivity. Consequently, the Confusion Matrix shows an outstanding recall rate: the system successfully detected 1,767 out of 1,813 yawns (97.5%). While this sensitive threshold did result in 409 false positives out of 519 normal images, this "safety-first" bias is intentional. Because we combine this AI data with the geometric Mouth Aspect Ratio (MAR) in our final live system, these AI-generated false positives are easily filtered out, leaving us with a highly accurate and robust yawn detection pipeline.

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================= FINAL SUMMARY ==================
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[EYE AUTOENCODER]
Best Threshold: 0.001899
              precision    recall  f1-score   support

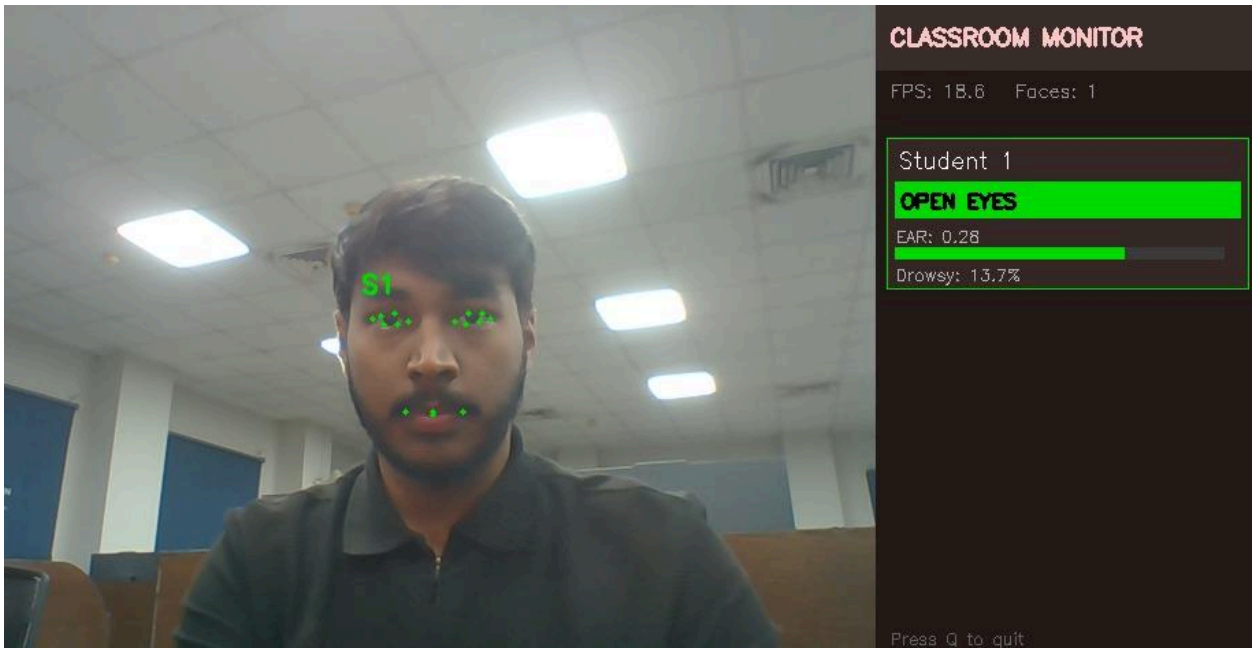
      Normal       0.00      0.00      0.00       400
     Anomaly       0.83      1.00      0.91      2000

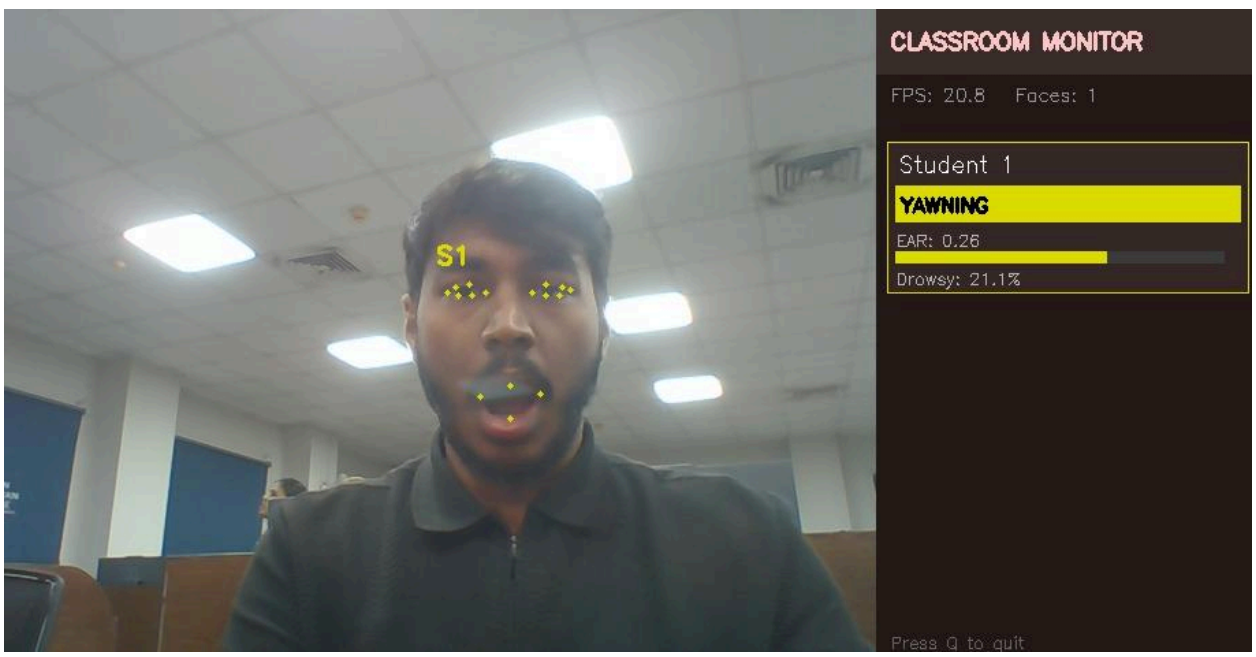
    accuracy                           0.83      2400
   macro avg       0.42      0.50      0.45      2400
weighted avg       0.69      0.83      0.76      2400


[YAWN AUTOENCODER]
Best Threshold: 0.019949
              precision    recall  f1-score   support
```

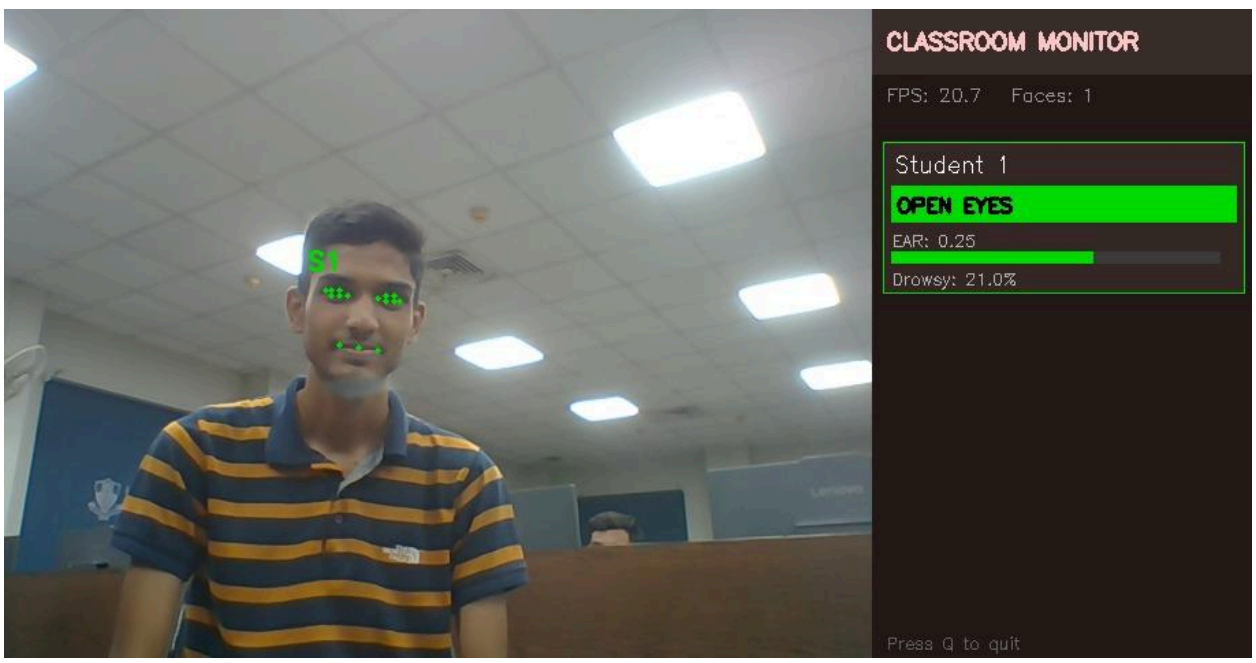
This summary report mathematically justifies our dual-condition system architecture. It proves that our Autoencoder is an incredibly powerful, highly sensitive trigger for detecting closed eyes (100% recall), but it also proves that we cannot rely on it alone. By taking this highly sensitive AI output and combining it with our geometric Eye Aspect Ratio (EAR) calculations in the live system, we successfully filter out the false positives, resulting in a robust, real-time drowsiness detector.

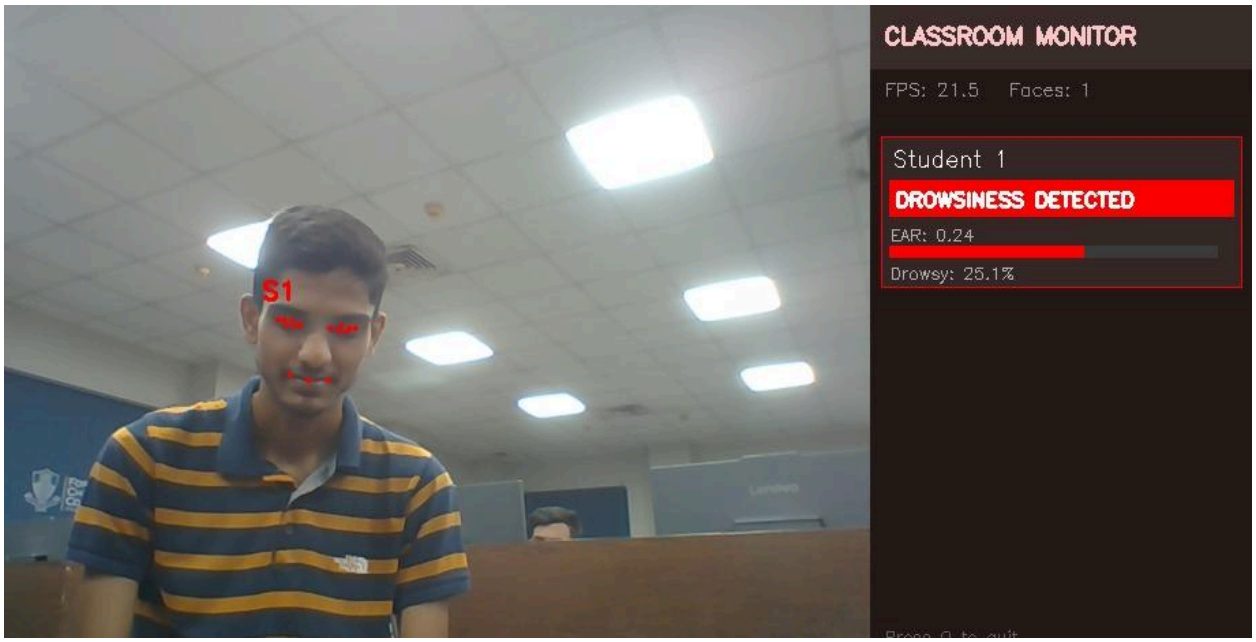
4. Testing

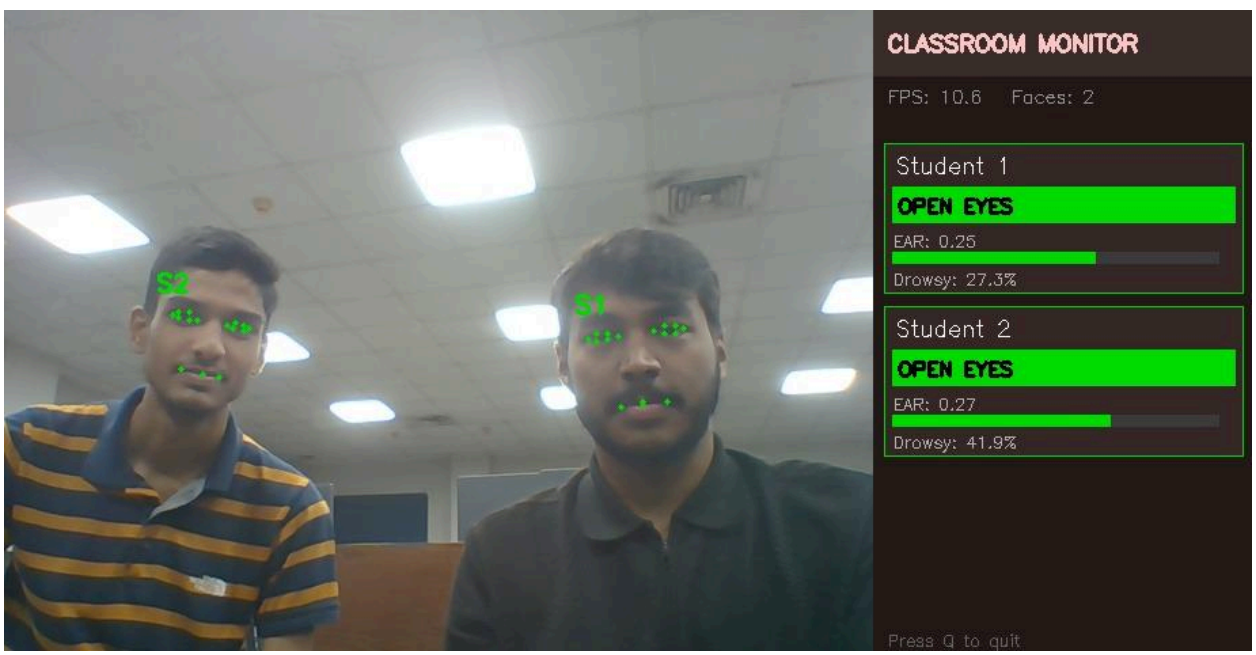

S1
CLASSROOM MONITOR
FPS: 18.6 Faces: 1
Student 1
OPEN EYES
EAR: 0.28
Drowsy: 13.7%
Press Q to quit

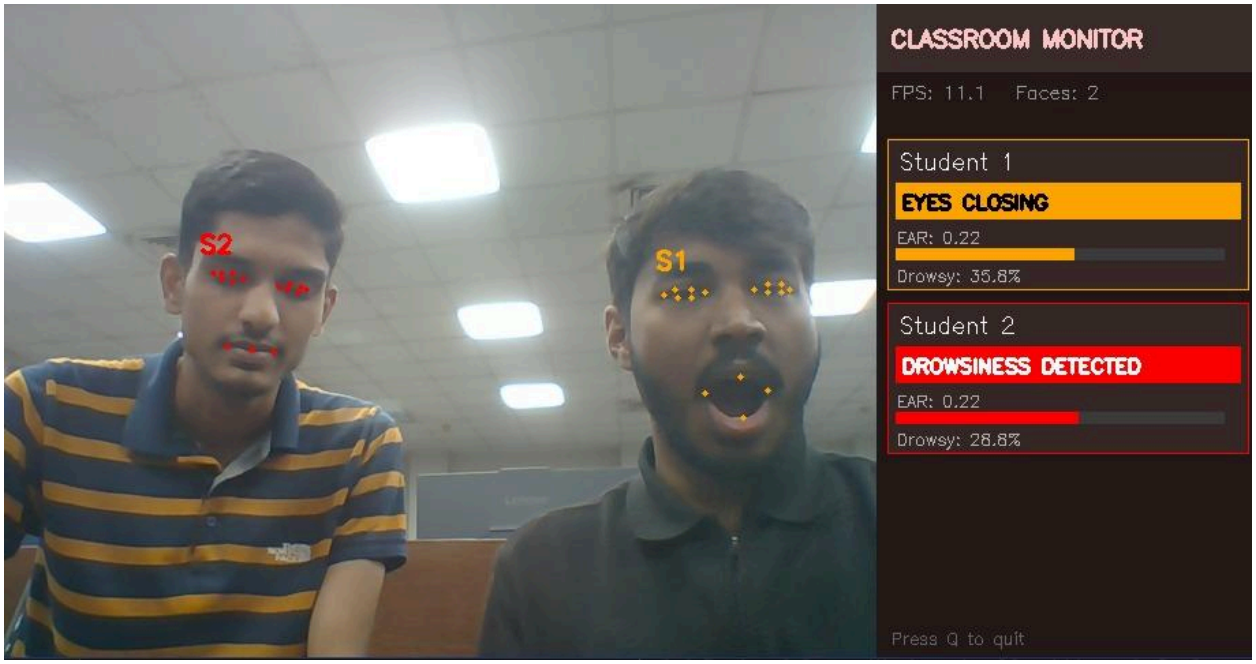

S1
CLASSROOM MONITOR
FPS: 20.8 Faces: 1
Student 1
YAWNING
EAR: 0.26
Drowsy: 21.1%
Press Q to quit

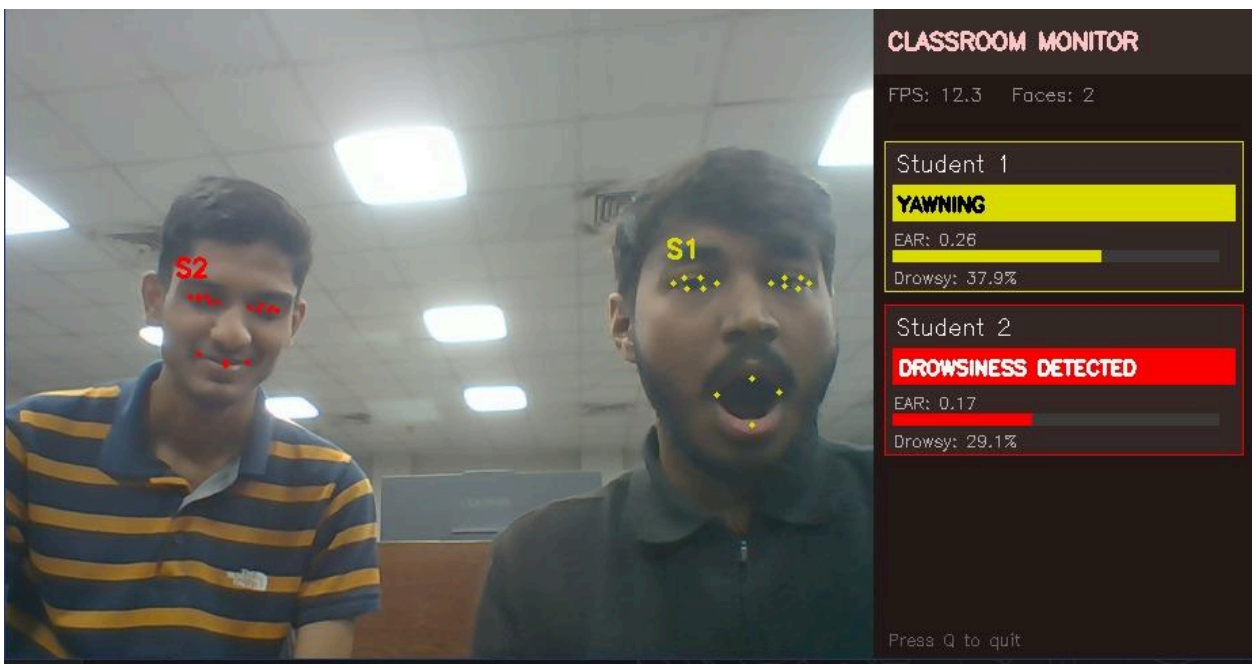
S1
CLASSROOM MONITOR
FPS: 21.1 Faces: 1
Student 1
EYES CLOSING
EAR: 0.22
Drowsy: 47.6%
Press Q to quit

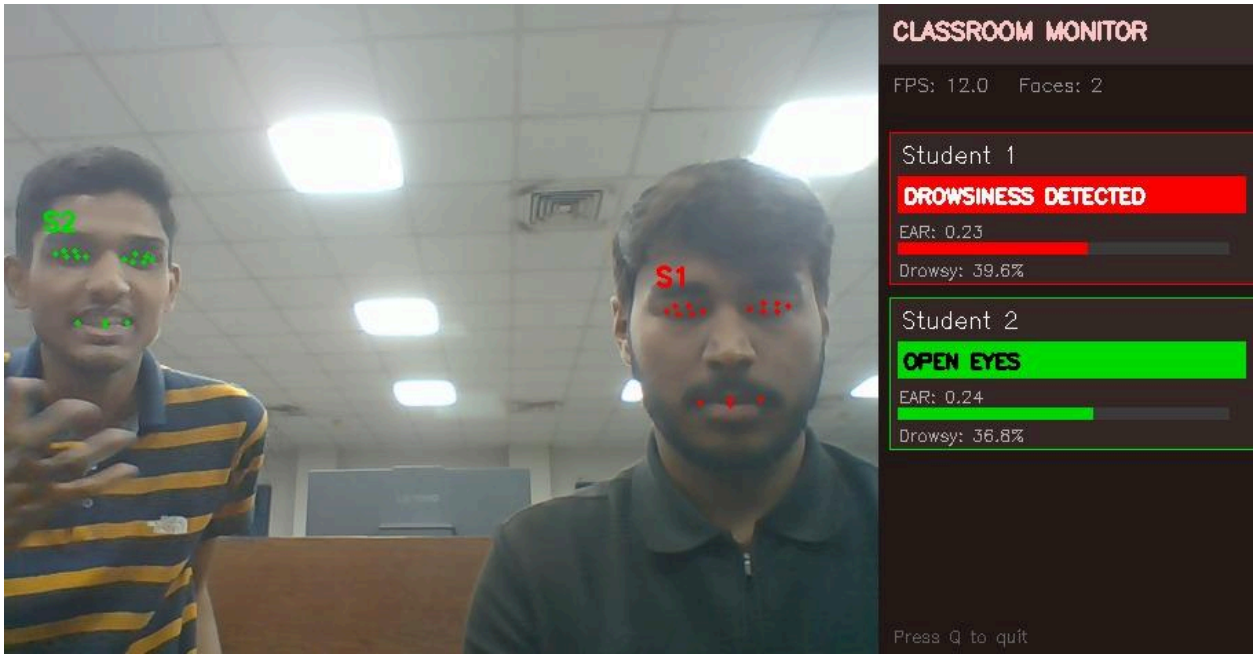

S1
CLASSROOM MONITOR
FPS: 20.7 Faces: 1
Student 1
OPEN EYES
EAR: 0.25
Drowsy: 21.0%
Press Q to quit

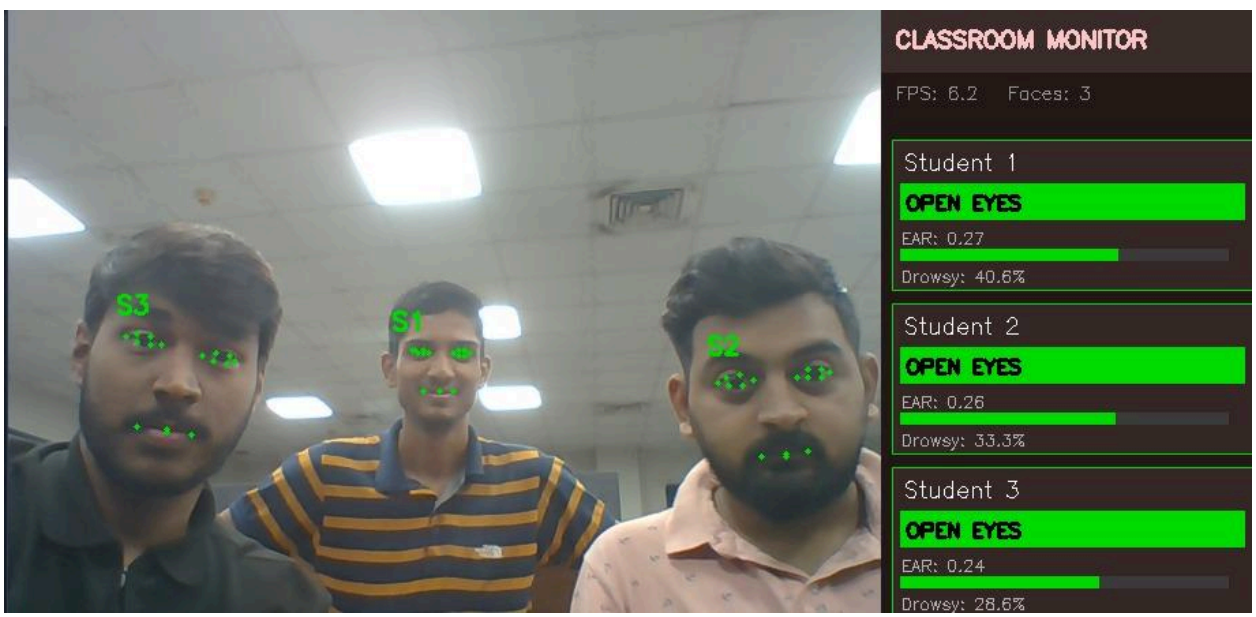

S1
CLASSROOM MONITOR
FPS: 21.5 Faces: 1
Student 1
DROWSINESS DETECTED
EAR: 0.24
Drowsy: 25.1%

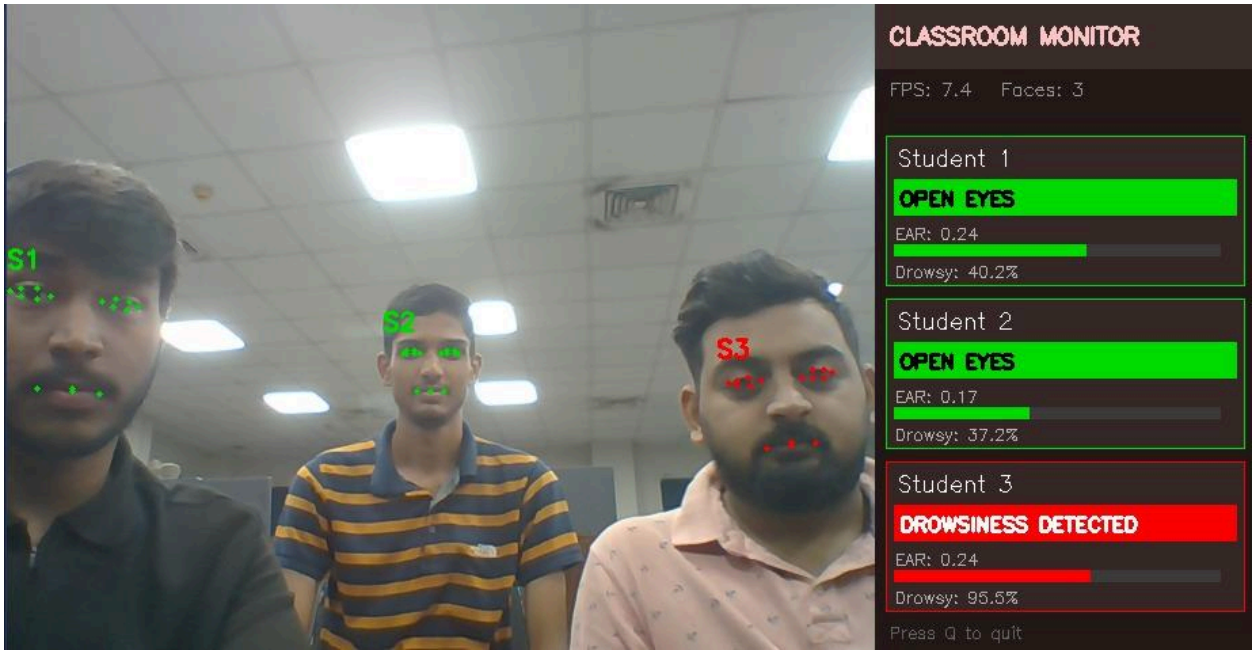

S2
S1
CLASSROOM MONITOR
FPS: 10.6 Faces: 2
Student 1
OPEN EYES
EAR: 0.25
Drowsy: 27.3%
Student 2
OPEN EYES
EAR: 0.27
Drowsy: 41.9%
Press Q to quit

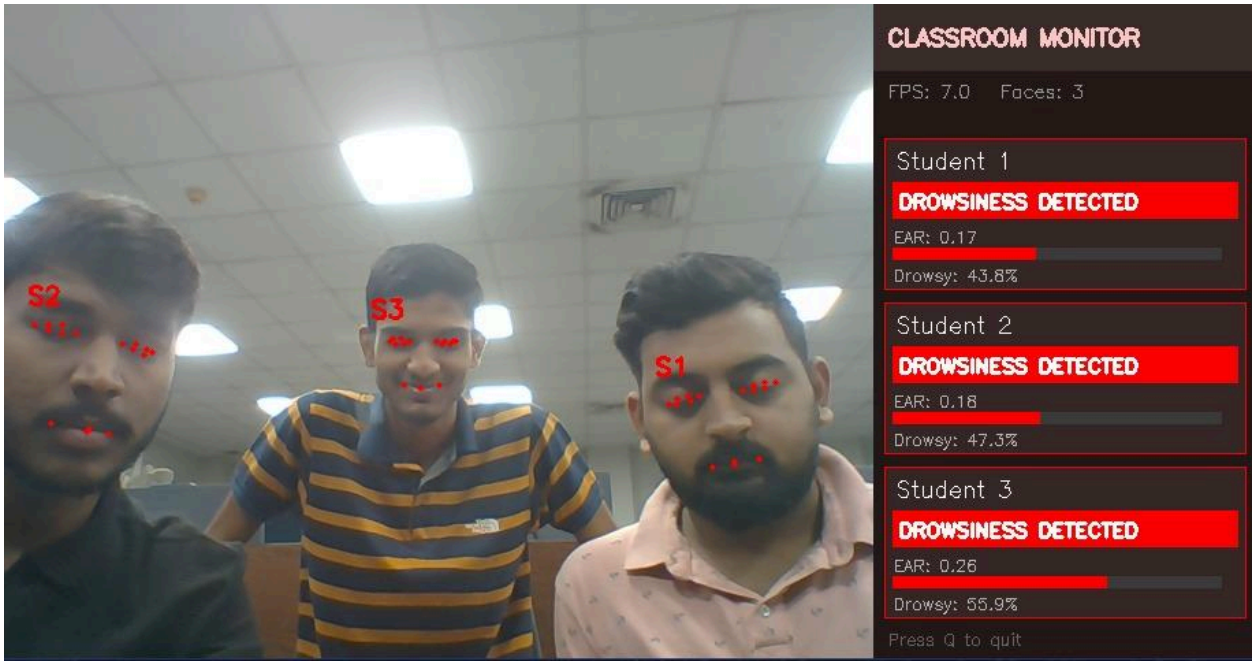

S2
S1
CLASSROOM MONITOR
FPS: 11.1 Faces: 2
Student 1
EYES CLOSING
EAR: 0.22
Drowsy: 35.8%
Student 2
DROWSINESS DETECTED
EAR: 0.22
Drowsy: 28.8%
Press Q to quit


S2
S1
CLASSROOM MONITOR
FPS: 12.3 Faces: 2
Student 1
YAWNING
EAR: 0.26
Drowsy: 37.9%
Student 2
DROWSINESS DETECTED
EAR: 0.17
Drowsy: 29.1%
Press Q to quit

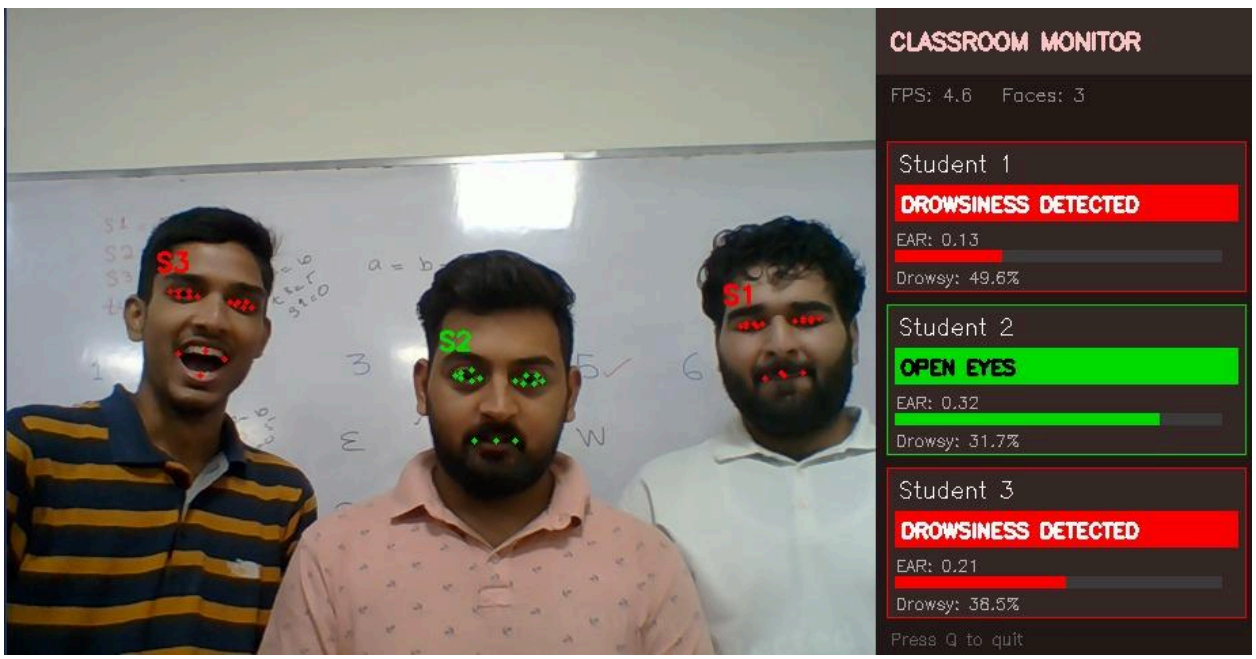

S2
S1
CLASSROOM MONITOR
FPS: 12.0 Faces: 2
Student 1
DROWSINESS DETECTED
EAR: 0.23
Drowsy: 39.6%
Student 2
OPEN EYES
EAR: 0.24
Drowsy: 36.8%
Press Q to quit


S3
S1
S2
CLASSROOM MONITOR
FPS: 6.2 Faces: 3
Student 1
OPEN EYES
EAR: 0.27
Drowsy: 40.6%
Student 2
OPEN EYES
EAR: 0.26
Drowsy: 33.3%
Student 3
OPEN EYES
EAR: 0.24
Drowsy: 28.6%


S1
S2
S3
CLASSROOM MONITOR
FPS: 7.4 Faces: 3
Student 1
OPEN EYES
EAR: 0.24
Drowsy: 40.2%
Student 2
OPEN EYES
EAR: 0.17
Drowsy: 37.2%
Student 3
DROWSINESS DETECTED
EAR: 0.24
Drowsy: 95.5%
Press Q to quit


S2
S3
S1
CLASSROOM MONITOR
FPS: 7.0 Faces: 3
Student 1
DROWSINESS DETECTED
EAR: 0.17
Drowsy: 43.8%
Student 2
DROWSINESS DETECTED
EAR: 0.18
Drowsy: 47.3%
Student 3
DROWSINESS DETECTED
EAR: 0.26
Drowsy: 55.9%
Press Q to quit

S1
CLASSROOM MONITOR
FPS: 15.1 Faces: 1
Student 1
DROWSINESS DETECTED
EAR: 0.23
Drowsy: 22.8%
Press Q to quit


S3
S2
S1
CLASSROOM MONITOR
FPS: 4.6 Faces: 3
Student 1
DROWSINESS DETECTED
EAR: 0.13
Drowsy: 49.6%
Student 2
OPEN EYES
EAR: 0.32
Drowsy: 31.7%
Student 3
DROWSINESS DETECTED
EAR: 0.21
Drowsy: 38.5%
Press Q to quit